



Parul University
Faculty of Engineering and Technology
Parul Institute of Engineering and Technology
Department of Artificial Intelligence and Data Science

Name of the teacher: Dr. Sandeep Wadekar
Subject Name: 303105306 | Theory of Computations

Hours/Week: 3
Hours
Mon, Tue, Thu

Sr. No.	Name of Topic	Planned Date
1	Alphabet, languages and grammars	08-06-2026
2	productions and derivation, Chomsky hierarchy of languages	09-06-2026
3	Introduction to Regular Expressions and Regular Languages	11-06-2026
4	Deterministic Finite Automata (DFA) – Definitions and Construction	15-06-2026
5	Equivalence of DFA and Regular Expressions	16-06-2026
6	Moore Machines – Concepts and Examples	18-06-2026
7	Mealy Machines – Concepts and Examples	22-06-2026
8	Conversion between Mealy and Moore Machines	23-06-2026
9	Nondeterministic Finite Automata (NFA) – Definitions and Examples	25-06-2026
10	Equivalence of NFA and DFA	29-06-2026
11	Regular Grammars and Equivalence with Finite Automata	30-06-2026
12	Properties of Regular Languages (Closure, Decision Properties)	02-07-2026
13	Pumping Lemma for Regular Languages – Statement and Applications	06-07-2026
14	Minimization of Finite Automata – Methods and Examples	07-07-2026
15	Introduction to Context-Free Grammars (CFG) and Context-Free Languages (CFL)	09-07-2026
16	Derivations and Parse Trees in CFG	13-07-2026
17	Ambiguity in CFG – Examples and Implications	14-07-2026
18	Chomsky Normal Form (CNF) – Definition and Conversion	16-07-2026
19	Greibach Normal Form (optional mention) – Basic Understanding (if time permits)	20-07-2026
20	Nondeterministic Pushdown Automata (PDA) – Introduction and Basics	21-07-2026
21	PDA Construction from CFG	23-07-2026

22	Equivalence of PDA and CFG	27-07-2026
23	Deterministic Pushdown Automata (DPDA) – Introduction and Examples	28-07-2026
24	Differences Between DPDA and NPDA	30-07-2026
25	Pumping Lemma for Context-Free Languages – Statement and Applications	03-08-2026
26	Closure Properties of Context-Free Languages	04-08-2026
27	Introduction to Context-Sensitive Languages (CSL)	06-08-2026
28	Context-Sensitive Grammars (CSG) – Definitions and Examples	10-08-2026
29	Comparison Between CFLs and CSLs + Summary & Practice Problems	11-08-2026
30	Basic Model of Turing Machines (TM) – Definition, Components, and Examples	13-08-2026
31	Turing-Recognizable (Recursively Enumerable) Languages	24-08-2026
32	Turing-Decidable (Recursive) Languages	25-08-2026
33	Closure Properties of Turing-Recognizable and Turing-Decidable Languages	27-08-2026
34	Variants of Turing Machines – Multi-tape, Multi-track, etc.	31-08-2026
35	Nondeterministic Turing Machines – Definition and Behavior	01-09-2026
36	Equivalence Between Deterministic and Nondeterministic Turing Machines	03-09-2026
37	Unrestricted Grammars – Definitions and Examples	07-09-2026
38	Equivalence of Unrestricted Grammars and Turing Machines	08-09-2026
39	Turing Machines as Enumerators – Concept and Examples	10-09-2026
40	Church-Turing Thesis – Statement, Significance, and Informal Arguments	14-09-2026
41	Formal Definition and Concept of the Universal Turing Machine (UTM)	17-09-2026
42	Construction and Working of a Universal Turing Machine	21-09-2026
43	Universal Language – Definition and Properties	22-09-2026
44	Diagonalization Language – Concept and Motivation	24-09-2026
45	Implications of Universal and Diagonalization Languages + Summary	28-09-2026